

2	(a)	(i)	Internal energy of a system is the sum of the random distribution of kinetic and potential energies associated with the molecules of the system	
		(ii)	There is no exchange of materials with the environment.	
		(iii)	Flow (transfer) of thermal energy (OR thermal energy in transit).	
	(b)	(i)	$W = 0$ since the gas is expanding into a vacuum and $Q = 0$ since container is insulated, therefore change in $U = 0$ Since U is proportional to T , change in $T = 0$	
		(ii)	$P_1 / P_2 = V_2 / V_1$ ratio = 3	
	(c)	(i)	A to B & D to A: Heat supplied to B to C & C to D: Heat supplied from	

		(ii)	$\Delta U = \frac{3}{2} nR \Delta T$ $\Delta U = \frac{3}{2} nR(2T) = 3nRT$	
		(iii)	$Q_{A \rightarrow B} = -nRT \ln \frac{V_i}{V_f} = nRT \ln \frac{V_f}{V_i} = nR(3T) \ln 2$ $Q_{C \rightarrow D} = -nRT \ln \frac{V_i}{V_f} = -nRT \ln 2$ For the other two process, the net heat supplied to system sums to be zero, so they can be excluded from calculations. $Q_{net} = 3nRT \ln 2 - nRT \ln 2 = 2 \ln 2 nRT = 1.38 nRT$	
		(iv)	$e = \frac{W}{Q}$ $e = \frac{2 \ln 2 nRT}{(3 \ln 2 + 3)nRT}$ $= 0.273$	

3	(c)			
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